

Positional Encoding in Transformers

A detailed, intuition-first set of notes covering position, dimensions, i , frequency, sine/cosine, collisions, and the original Transformer formula.

Core idea: Self-attention can compare every token with every other token, but the original attention mechanism does not automatically know whether a token came first, second, third, and so on. Positional encoding injects sequence-order information into the representations.

Learning goal

- Understand exactly what pos , i , $2i$, $2i+1$, and d_{model} mean.
- Understand frequency as the rate at which a sinusoidal signal changes as position increases.
- See numerically how a positional encoding is calculated for a 5-dimensional vector.
- Understand why sine alone can encode position, but has ambiguity because it is periodic and not one-to-one.
- Understand why sine and cosine are used together.
- Understand why multiple frequencies are used.
- Understand why tangent is not useful as an additional positional signal.
- Understand what happens when positional encoding is added to a word embedding.

1. Why do Transformers need positional information?

A token embedding tells the model something about the token itself, but the original self-attention mechanism does not inherently attach a built-in sequence index to each token. This matters because word order changes meaning.

Example: "Dog bites man" vs. "Man bites dog"

The same vocabulary items appear, but their order is different. A Transformer therefore needs an explicit way to represent position.

Important distinction: positional encoding does not replace the token embedding. It is combined with the token embedding so the resulting representation contains both token information and positional information.

Transformer input representation = token embedding + positional encoding

Conceptually:

Tokens → Token IDs → Embeddings → ADD positional information → Transformer layers

2. The two different counters: pos and i

This is one of the most important sources of confusion. The symbols **pos** and **i** mean completely different things.

pos means the position of the token in the sequence. For a sequence of length 5, positions might be 0, 1, 2, 3, 4.

i is a dimension-pair counter used while constructing the positional encoding vector. You do not derive **i** from the word. You simply iterate through **i** = 0, 1, 2, ... while filling the dimensions.

pos = which token position am I encoding? **i** = which sine/cosine frequency pair am I currently calculating?

For $d_{\text{model}} = 5$, the positional encoding has five dimensions:

Vector dimension	0	1	2	3	4
Function	sin	cos	sin	cos	sin
Pair counter i	0	0	1	1	2
Frequency denominator	1	1	$10000^{(2/5)}$	$10000^{(2/5)}$	$10000^{(4/5)}$

Memory trick: **pos** is the word location. **i** is the frequency/dimension-pair counter.

3. The original Transformer sinusoidal formula

The original Transformer used a fixed, deterministic sinusoidal positional encoding. For an embedding size d_{model} , the two formulas are:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\text{pos} / 10000^{(2i / d_{\text{model}})}\right)$$

$$\text{PE}(\text{pos}, 2i+1) = \cos\left(\text{pos} / 10000^{(2i / d_{\text{model}})}\right)$$

Interpretation:

- $2i$ selects an even dimension: 0, 2, 4, ...
- $2i+1$ selects the following odd dimension: 1, 3, 5, ...
- The same denominator is used for the sine/cosine pair.
- As i increases, the denominator increases, so the signal changes more slowly with position.
- The positional encoding is a vector with the same dimensionality as the token embedding.

4. What exactly is frequency here?

Frequency means how quickly a repeating signal changes or completes cycles as its input moves forward. In positional encoding, position is the input that moves forward: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots$

Imagine a light blinking. A light that turns on and off every 2 seconds has a higher frequency than one that turns on and off every 10 seconds. For sinusoidal encoding, the analogous idea is how quickly the sine/cosine value changes as position increases.

`sin(position) → changes relatively quickly`

`sin(position / 10) → changes more slowly`

`sin(position / 100) → changes even more slowly`

Key point: The denominator controls the speed of change. A larger denominator means the argument of sine/cosine changes more slowly as pos increases, producing a longer period and therefore a lower frequency.

5. Period and frequency

For a standard sine wave $\sin(x)$, one complete cycle takes 2π radians. This is its period in x . If we use $\sin(kx)$, the period becomes $2\pi/k$. Therefore, increasing k makes the wave repeat faster; decreasing k makes it repeat more slowly.

`sin(k · pos): period =  $2\pi / k$`

In the Transformer formula, the signal looks like $\sin(\text{pos} / D)$. This is equivalent to $\sin((1/D) \cdot \text{pos})$. Therefore $k = 1/D$ and the period is:

`period =  $2\pi D$`

So a larger denominator D gives a longer period. This is exactly why the later dimensions act like slower clocks.

6. A complete numerical example with $d_{\text{model}} = 5$

Suppose a word has the embedding vector:

`X = [0.2, 0.5, 0.1, 0.7, 0.9]`

Assume the word occurs at position $\text{pos} = 3$. The values in X are not used to calculate the positional encoding. They are used later when the positional encoding is added to the embedding.

Because $d_{\text{model}} = 5$, we need five positional values: dimensions 0 through 4.

Step 1 — $i = 0$

For the first pair, $2i = 0$ and $2i+1 = 1$.

$$PE(3,0) = \sin(3 / 10000^0) = \sin(3) \approx 0.1411$$

$$PE(3,1) = \cos(3 / 10000^0) = \cos(3) \approx -0.9900$$

For $i = 0$, the denominator is 1. This is the fastest-changing pair in this 5-dimensional example.

Step 2 — $i = 1$

Now we move to dimensions 2 and 3.

$$10000^{(2/5)} \approx 39.8107$$

$$PE(3,2) = \sin(3 / 39.8107) \approx 0.0753$$

$$PE(3,3) = \cos(3 / 39.8107) \approx 0.9972$$

The denominator is now about 39.8, so the signal changes much more slowly from one position to the next.

Step 3 — $i = 2$

We now reach dimension 4. Since $d_model = 5$, there is no dimension 5; the vector stops at dimension 4.

$$10000^{(4/5)} \approx 1584.893$$

$$PE(3,4) = \sin(3 / 1584.893) \approx 0.0019$$

Final positional encoding

$$PE(3) \approx [0.1411, -0.9900, 0.0753, 0.9972, 0.0019]$$

7. Add the positional encoding to the word embedding

Now we combine the word representation X with the positional representation $PE(3)$:

$$X = [0.2, 0.5, 0.1, 0.7, 0.9]$$

$$PE(3) = [0.1411, -0.9900, 0.0753, 0.9972, 0.0019]$$

$$X + PE = [0.3411, -0.4900, 0.1753, 1.6972, 0.9019]$$

The resulting vector is what is passed onward. It now carries information about both **what the token is** and **where it occurs**.

8. Why use sine at all?

Sine is a smooth, continuous, deterministic function that changes systematically with position. If we define a positional signal as $\sin(\text{pos})$, then each position receives a numerical value.

Position 0 $\rightarrow \sin(0) = 0$

Position 1 $\rightarrow \sin(1) \approx 0.8415$

Position 2 $\rightarrow \sin(2) \approx 0.9093$

Position 3 $\rightarrow \sin(3) \approx 0.1411$

Therefore, sine alone can absolutely provide positional information. It is incorrect to say that cosine is required for positional encoding to work at all.

Important: Sine alone can encode position. The reason for adding cosine is that sine by itself is ambiguous within a cycle and provides only one coordinate of the phase.

9. The collision problem with sine

Sine is periodic. That means it repeats. The identity is:

$$\sin(x + 2\pi k) = \sin(x), \text{ for any integer } k$$

So if we allowed arbitrary real-valued positions, two different positions could have exactly the same sine value.

$$\sin(0) = 0$$

$$\sin(2\pi) = 0$$

$$\sin(4\pi) = 0$$

Therefore, with the deliberately chosen toy encoding $\text{PE}(\text{pos}) = \sin(\text{pos})$, positions 0, 2π , and 4π collide exactly.

In practical Transformer sequences, pos is an integer and the original frequency schedule is not simply $\sin(\text{pos})$ repeated with an integer period. So the above is a conceptual demonstration of periodic ambiguity, not a claim that the standard Transformer has obvious integer-position collisions at normal sequence lengths.

10. An integer-position toy example with an obvious collision

To make the collision easy to see using integer positions, choose a toy signal:

$$\text{PE}(\text{pos}) = \sin(\pi \cdot \text{pos} / 2)$$

Now calculate:

Position	Angle	$\sin(\text{angle})$
0	0	0
1	$\pi/2$	1
2	π	0
3	$3\pi/2$	-1

4	2π	0
5	$5\pi/2$	1

Collision: positions 0, 2, and 4 all produce the value 0. Sine alone does not give a one-to-one mapping from position to value.

11. Why add cosine?

For one frequency, sine gives one coordinate of a rotating point. Cosine gives the other coordinate. Together they form the pair:

$(\cos(\theta), \sin(\theta))$

On a unit circle, cosine is the horizontal coordinate and sine is the vertical coordinate. Together they identify the point around the circle much more completely than sine alone.

$\sin(\theta) = 0.5$ can occur at $\theta = \pi/6$ and $\theta = 5\pi/6$

At $\theta = \pi/6$: $[\sin \theta, \cos \theta] = [0.5, 0.866]$

At $\theta = 5\pi/6$: $[\sin \theta, \cos \theta] = [0.5, -0.866]$

The sine values are identical, but the cosine values are different. Therefore the pair distinguishes the two phases.

Intuition: If I tell you only the vertical height of a point on a circle, there can be two possible locations. If I tell you both horizontal and vertical coordinates, I know the location.

12. Does sine + cosine completely eliminate collisions?

No. This distinction is important. A single sine/cosine pair is still periodic.

$[\sin(\theta), \cos(\theta)] = [\sin(\theta + 2\pi), \cos(\theta + 2\pi)]$

So positions separated by a full cycle can still have the same pair if the frequency is fixed.

The original Transformer addresses this by using **many sine/cosine pairs at different frequencies**. The complete positional encoding is therefore a high-dimensional combination of periodic signals.

13. Why multiple frequencies?

Imagine several clocks. One clock completes a cycle quickly; another completes a cycle more slowly; another is much slower. At any position, you read all the clocks at once.

Fast signal → changes a lot from position to position

Medium signal → changes moderately

Slow signal → changes only a little

One signal may repeat and therefore give the same value at two positions. But another signal with a different period may give a different value at those same positions.

That combination creates a positional fingerprint.

14. The role of i revisited: i chooses the frequency

Now the meaning of i should be clearer. i is not the word position. It indexes the sine/cosine frequency pair.

$i = 0 \rightarrow$ dimensions 0 and 1 $\rightarrow$ first, fastest pair

$i = 1 \rightarrow$ dimensions 2 and 3 $\rightarrow$ second, slower pair

$i = 2 \rightarrow$ dimensions 4 and 5 $\rightarrow$ third, slower pair

For an odd d_{model} such as 5, the final pair is incomplete because the vector ends at dimension 4. That is why our example had a sine at dimension 4 but no corresponding cosine at dimension 5.

15. Why does the Transformer use $10000^{(2i/d_{\text{model}})}$?

The expression $10000^{(2i/d_{\text{model}})}$ creates a geometric progression of scale values across dimensions. As i grows, the denominator grows smoothly, giving progressively slower-changing signals.

$$D_i = 10000^{(2i/d_{\text{model}})}$$

$$\text{signal}_i(\text{pos}) = \sin(\text{pos} / D_i) \text{ or } \cos(\text{pos} / D_i)$$

The important idea is not to memorize the number 10000 as if it were a magic constant. It is a design choice from the original Transformer formulation that creates a wide range of wavelengths.

For $d_{\text{model}} = 5$, the denominators we calculated were approximately:

i	Dimensions	D_i	Behavior
0	0, 1	1	Fastest
1	2, 3	39.81	Slower
2	4 (and 5 if present)	1584.89	Very slow

As a result, the positional encoding contains both fine-grained signals and long-range signals. The fast signals can change noticeably between neighboring positions, while the slow signals preserve a slowly varying notion of where we are in the sequence.

16. A second numerical comparison: position 3 vs. position 4

Using the same $d_{\text{model}} = 5$ setup:

$$\text{PE}(3) \approx [0.1411, -0.9900, 0.0753, 0.9972, 0.0019]$$

$$\text{PE}(4) \approx [-0.7568, -0.6536, 0.1004, 0.9950, 0.0025]$$

Look at the first pair:

$$\text{Position 3} \rightarrow [0.1411, -0.9900]$$

$$\text{Position 4} \rightarrow [-0.7568, -0.6536]$$

There is a large change. This is the fast signal.

Now look at the second pair:

$$\text{Position 3} \rightarrow [0.0753, 0.9972]$$

$$\text{Position 4} \rightarrow [0.1004, 0.9950]$$

The change is much smaller. This is the slower signal.

Finally, the very slow dimension changes only slightly:

$$\text{Position 3} \rightarrow 0.0019$$

$$\text{Position 4} \rightarrow 0.0025$$

This is the practical meaning of multiple frequencies: different dimensions react to movement through the sequence at different speeds.

The positional vector is not saying "the position is the number 3." Instead, it is saying something more like: "At position 3, each of my many sinusoidal clocks is at this particular phase."

21. A very important correction about 'never collide'

It is tempting to say: "Sine and cosine with multiple frequencies guarantee that different positions can never collide." That is too strong as a general mathematical statement.

The accurate statement is:

The original sinusoidal scheme uses multiple sine/cosine signals with different wavelengths so that positions are represented by a rich, high-dimensional, multi-scale pattern. This makes positions distinguishable over practical sequence lengths and provides useful relative-position structure. Individual signals are periodic, and a single pair can certainly repeat.

Do not memorize the false rule "sine and cosine never collide." Memorize the mechanism: **multiple bounded periodic signals with different scales create a high-dimensional positional fingerprint.**

22. A compact derivation for $d_{\text{model}} = 5$

If $d_{\text{model}} = 5$ and we want position $\text{pos} = p$, then the complete vector is obtained by taking $i = 0, 1, 2$ and filling the valid dimensions.

```
Dimension 0:  $\text{PE}(p, 0) = \sin(p / 10000^0)$ 
Dimension 1:  $\text{PE}(p, 1) = \cos(p / 10000^0)$ 
Dimension 2:  $\text{PE}(p, 2) = \sin(p / 10000^{(2/5)})$ 
Dimension 3:  $\text{PE}(p, 3) = \cos(p / 10000^{(2/5)})$ 
Dimension 4:  $\text{PE}(p, 4) = \sin(p / 10000^{(4/5)})$ 
```

Notice that i appears in the formula only to generate the dimension indices and the scale for that pair. You can implement this as a loop over i , but conceptually it is simply a bookkeeping index.

23. Pseudocode

```
for i = 0, 1, 2, ... denominator = 10000 ** (2*i / d_model) even_dimension = 2*i
PE[pos, even_dimension] = sin(pos / denominator) odd_dimension = 2*i + 1 if
odd_dimension < d_model: PE[pos, odd_dimension] = cos(pos / denominator)
```

For $d_{\text{model}} = 5$, the loop visits $i = 0, 1, 2$. The last iteration writes dimension 4 with sine; dimension 5 would be outside the vector, so no cosine value is written for it.

24. Interview-ready explanation

Answer: "Transformers use positional encoding because self-attention does not inherently encode the order of tokens. In the original Transformer, positional information is added to token embeddings using sinusoidal functions. Even dimensions use sine and odd dimensions use cosine. The index pos is the token position, while i is the index of the sine/cosine frequency pair. As i increases, the denominator $10000^{(2i/d_{\text{model}})}$ increases, so the sinusoidal signal changes more slowly, giving different dimensions different wavelengths. Sine alone could encode position, but because it is periodic it has ambiguity within a cycle. Adding cosine gives the second coordinate of the phase, and using many frequencies creates a high-dimensional positional fingerprint."

25. Common misconceptions

- **Misconception:** i is the token position. **Correction:** pos is the token position; i indexes the frequency/dimension pair.
- **Misconception:** sine cannot encode position by itself. **Correction:** sine can encode position, but periodicity creates ambiguity.
- **Misconception:** sine + cosine can never collide. **Correction:** a single pair is still periodic; multiple frequencies provide the useful multi-scale representation.
- **Misconception:** frequency is an extra number stored in the vector. **Correction:** frequency describes how quickly the sinusoidal value changes with position.
- **Misconception:** tangent should be added because it is another trig function. **Correction:** tangent is derived from sine/cosine, unbounded, and undefined at cosine = 0.

• **Misconception:** positional encoding depends on the word's embedding values. **Correction:** the original fixed sinusoidal encoding depends on position and d_{model} , not the semantic content of the word.

26. Final mental model

TOKEN EMBEDDING = what the token means POSITIONAL ENCODING = where the token is
pos = token position i = which frequency pair / dimensions frequency = how quickly a
sinusoidal signal changes with position sine = one smooth periodic signal cosine =
complementary phase coordinate many frequencies = many clocks at different speeds
final input = token embedding + positional encoding

If you remember only one sentence: Positional encoding represents each token position as a high-dimensional fingerprint made from sine/cosine waves that change at different speeds, and this fingerprint is added to the token embedding.